

UA Little Rock Information Quality Graduate Program: Overview

**UALR Donaghey College of
Science, Technology, Engineering,
and Mathematics (CSTEM)**

**Dr. John Talburt
Graduate Coordinator**



What is Information Quality?

- Many Definitions
 - Accuracy
 - Fitness for use
 - Accurate, complete, and timely information that is free from defect
 - Right information at the right time, at the right place, for the right people
 - The application of Total Quality Management principles to building Information Products

UA Little Rock Definition of IQ

- Information Quality is an Information Science discipline concerned with
 - **Maximizing the value** and minimizing the risk of an organization's information assets and
 - Assuring the information products produced by the organization **create value for the people and systems** who use them

IQ Knowledge & Practice

- Information Quality is a discipline, a body of knowledge and practice covering
 - Information quality value & business impact
 - Information quality strategy & governance
 - Information quality environment & culture
 - Information quality measurement & improvement
 - Sustaining information quality
 - Information architecture quality
 - *IQ International six domains of IQ

**Data
Quality**

Why Study Information Quality?

- It is important!
Poor IQ
 - Cost billions of dollars each year
 - Costs lives
- The rise of Generative AI has given it new focus!
 - Organizations are realizing their AI models are only as good as the data used to train them
 - Agentic AI is a way to use AI to improve IQ
- YOU can make a difference!

We Are a Community

- More than just students
 - We support each other
 - We respect each other especially professors, teaching assistants, senior class members, graduates
 - Take advantage to
 - Build your network
 - Build your professional credentials
 - Gain professional experience
- LinkedIn “UALR Information Quality Graduate Program”**

Information Science Department

- Separated from Computer and Information Science in 1999
 - Dr. Elizabeth Pierce, Chair
- Focuses on data and information aspects of computing
- Graduate Programs
 - Information Quality GC, MS, PhD
 - Information Science MS, PhD
 - Data Science GC
 - Bioinformatics MS, PhD

History of the IQ Program

- MSIQ Admissions Opened – May 10, 2006
- First fall class of 25 students – Fall 2006
- Live webcast of MSIQ classes started – Fall 2007
- IQ Track in Applied Science PhD – Spring 2008
- Certificate Program started – Spring 2008
- IQ Track moves to Integrated Comp PhD – Fall 2010
- Name Change: Computer and Information Sciences PhD – May 2017
- PhD Program approved for Online – Fall 2017

Founding Sponsors

- **Acxiom Corporation**, Little Rock, AR
Charles Morgan, Company Leader
- **MIT IQ Program**, Cambridge, MA
Dr. Richard Wang, Director
- **UALR Donaghey College of Engineering and Information Technology**
Dr. Mary Good, Founding Dean

Program Purpose & Objectives

A curriculum designed to meet the growing demands by government and industry for qualified professionals with a graduate-level degree that encompasses

- Concepts, principles, tools, and models essential in defining, measuring, analyzing, and improving information quality

Program Objectives (Cont.)

- Information quality policies, strategies, programs, and standards (e.g. ISO 8000, ISO 25500)
- The role of information quality in operational improvement, data privacy and protection, risk management, data governance, information architecture, data integration, and master data management

Degrees Offered

- Master of Science in Information Quality
- Graduate Certificate in Information Quality
- Computer and Information Sciences PhD - Information Quality Track

MSIQ Curriculum (33 hours)

<u>Information Quality</u>	<u>Information Science</u>
Principles (7303)	Info Visualization (5345)
Policy & Strategy (7367)	Info Sys Analysis (7310)
IQ Tools (7342)	Database Sys (7320)
Proj & Chg Mgmt (7337)	
INFQ Elective	IFSC Elective
Project, Thesis, or 2 more Electives (6 hrs)	

Graduate Certificate Program

- First 12 hours of the MSIQ program
- No GRE required for admission
- Completion with 3.5+ GPA waives GRE for admission to MSIQ
- Courses
 - INFQ 7303 Principles of IQ
 - INFQ 7367 IQ Policy and Strategy
 - INFQ 7342 IQ Tools & Industry Landscape
 - Elective with INFQ prefix, e.g. INFQ 7322, INFQ 7337, INFQ 7348,...

Graduate Certificate (12 hr)

<u>Information Quality</u>	<u>Information Science</u>
Principles (7303)	Info Visualization (5345)
Policy & Strategy (7367)	Info Sys Analysis (7310)
IQ Tools (7342)	Database Sys (7320)
Prj & Chg Mgmt (7337)	
or	
INFQ Elective	IFSC Elective
Thesis or Project	

Every MSIQ & PhD-IQ Student should get GC IQ!

- If in MSIQ or IQ PhD, apply for GC in IQ now!
 - For IQ GC, no extra courses to take – All courses for IQ GC are part of the MSIQ
 - If you want the Data Science GC, you will have to take one extra course. DS GC requires
 - IFSC 7320 (req by MSIQ)
 - IFSC 5345 (req by MSIQ)
 - IFSC 7360
 - IFSC 7370
- Only One Counts toward MSIQ

PhD Course Requirements (75 hours)

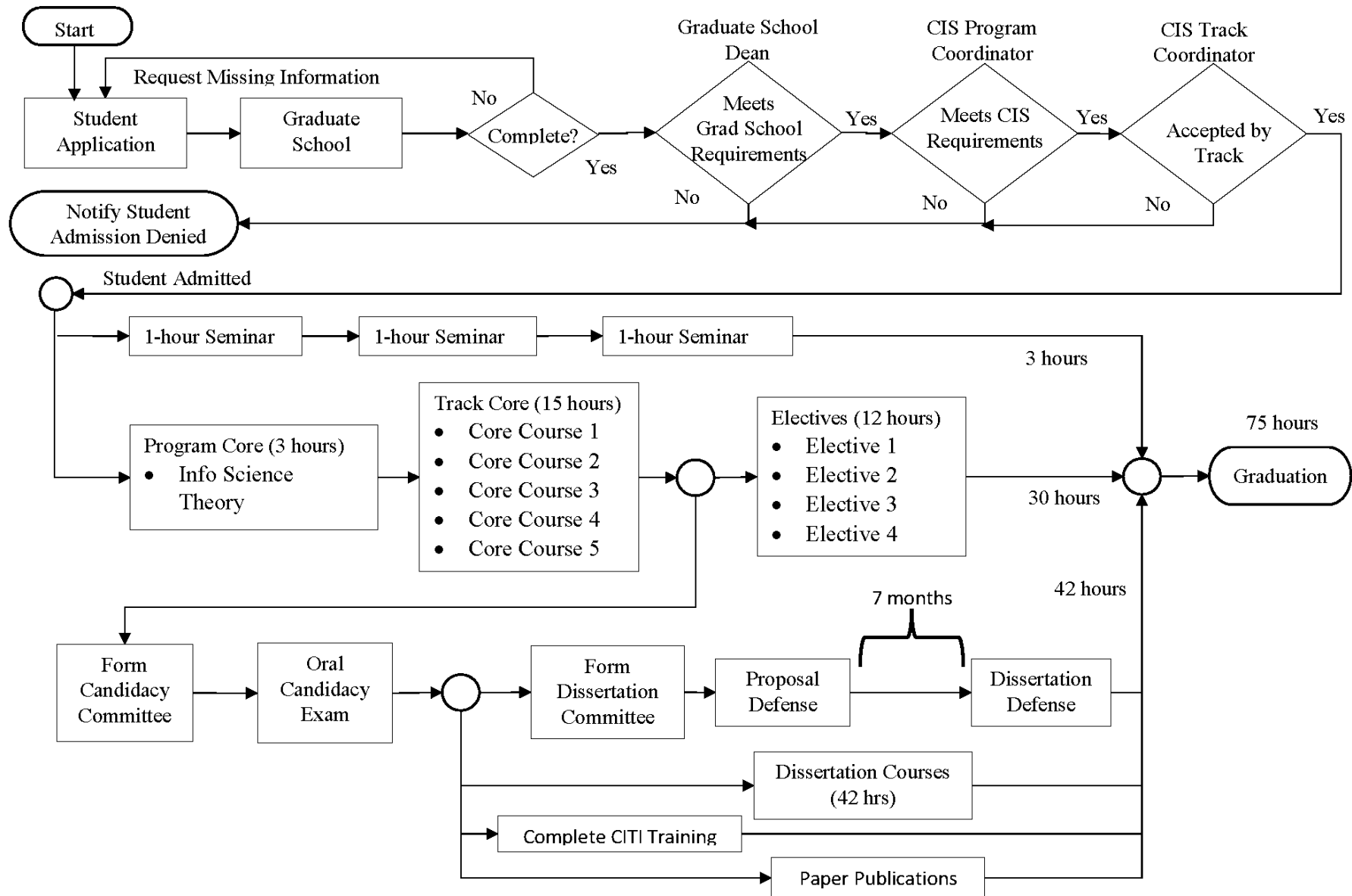
- Lecture Classes 33 hours
 - Program Core (6 hours):
 - IFSC 7321, Info Thy
 - IFSC 7310, Sys Analysis
 - Track Core (12 hours):
 - INFQ 7303, IQ Principles
 - IFSC 7320, Database
 - INFQ 7322, IQ Theory
 - INFQ 7367, IQ Policy & Strategy
 - Four electives (12 hours) & 3 seminars (3 hours)
- Dissertation research 42 hours

PhD Milestones

- Candidacy Exam
 - At or near completions of 18-hour core
 - Must have Dissertation Advisor
 - Oral exam with 3 other IS or CS faculty members
- Complete online CITI Training (submit certificate)
- Select Dissertation Committee
 - Advisor + 3 other faculty (One can be external/affiliate)
- Defend Research Proposal
 - Complete Research and Write Dissertation
- Defend Research and Dissertation

Other Requirements

- Paper Publication
 - Must be 1st or 2nd author of a research paper related to your dissertation research
- Minimum Time between Proposal and Defense
 - Research proposal meeting and final dissertation defense meeting must be separated by at least 7 months (210 days)
- Consecutive Enrollment
 - Must register for at least one hour every Fall and Spring
 - Includes Summer after completing 42 dissertation hours



Computer and Information Sciences (CIS) PhD Program Process Flow

Full-Time Information Science Faculty

- Dr. Nitin Agarwal, (Social Comp, Dbase, Sys Analysis)
- Dr. Dan Berleant, (Info Thy, Sys Analysis)
- Dr. Serhan Dagtas, (Research)
- Dr. John Talburt (Prin IQ, Proj Mgmt, Entity Res)
- Dr. Edi Tudoreanu (Info Visualization)
- Dr. Ningning Wu (Data Mining, Network Sec)
- Dr. Xiaowei Xu (Info They, Dbase Sys, Adv Data Mining)
- Dr. Mary Yang (Bioinformatics)
- Dr. Skyler Thompson (IQ Policy & Strategy)
- Dr. Ahmed Abu-Halimeh (IQ Tools)
- Many Affiliate Faculty

Three Types of Course Sections

- On-Campus Sections
 - Required Classroom Attendance (01,02, ...)
 - Webcast & Some Classroom Attendance (9W1, 9W2, ...)
 - Hybrid Attendance as Noted in Schedule (H01, H02, ...)
- Online Campus
 - Fully Online (9U1, 9U2, ...)
- Must register for sections according to your campus designation

IQ Program LinkedIn Group

- “UALR Information Quality Graduate Program”
 - Open Group

PLEASE JOIN!!

- Announcements
- News
- Job postings
- Internship opportunities
- Members are current students, faculty, graduates, employers, and other stakeholders

Official Website

- UALR IQ website [//ualr.edu/informationquality](http://ualr.edu/informationquality)
 - News
 - Program
 - Admission requirements
 - Curriculum requirements
 - FAQ
 - Policies
 - Resource links

International Association for Data Quality, Governance, and Analytics (IADQGA)

- New Professional Organization
 - <https://www.iadqga.com/>
 - Free Membership!
 - Just Starting
 - Planning to offer a number resources
 - Focus on Data Literacy and Statistical Literacy

Questions and Discussion

